

OSTROV CHETYREHSTOLBOVOY, Russian Federation

WMO: 219650

Lat: 70.633N

Lon: 162.483E

Elev: 41

StdP: 100.83

Time Zone: 11.00 (E11)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.0	(c) -38.5	(d) -42.0	(e) 0.1	(f) -39.9	(g) -40.5	(h) 0.1	(i) -38.4	(j) 16.4	(k) -24.3	(l) 15.1	(m) -24.4	(n) 4.6	(o) 210	(p) 1.179

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.0	(c) 11.1	(d) 8.8	(e) 8.7	(f) 7.0	(g) 6.8	(h) 5.2	(i) 9.1	(j) 10.8	(k) 7.1	(l) 8.7	(m) 5.3	(n) 6.6	(o) 3.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 7.7	(b) 6.6	(c) 10.5	(d) 5.8	(e) 5.7	(f) 8.1	(g) 4.0	(h) 5.1	(i) 5.8	(j) 27.6	(k) 10.9	(l) 23.2	(m) 8.8	(n) 19.4	(o) 6.7	(p) 15.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	12.7	11.3	-41.0	14.9	1.4	3.4	-42.0	17.4	-42.9	19.4	-43.7	21.3	-44.7	23.7
			-41.0	11.6	1.4	2.7	-42.0	13.5	-42.9	15.1	-43.6	16.6	-44.6	18.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.1	-28.9	-27.2	-26.4	-17.6	-6.3	0.8	3.0	2.4	-0.3	-9.0	-20.2	-27.9		
	DBStd	13.28	5.71	6.47	5.99	5.42	4.51	2.50	2.49	2.82	2.45	5.14	6.07	5.91		
	HDD10.0	8417	1206	1042	1128	828	506	277	218	236	308	588	906	1173		
	HDD18.3	11457	1464	1275	1386	1078	764	527	476	493	558	846	1156	1432		
	CDD10.0	1	0	0	0	0	0	0	1	1	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.7	5.6	5.4	5.2	5.4	5.7	5.7	5.7	6.0	6.5	5.9	6.0	5.4		
(15) Precipitation	PrecAvg	88	3	2	2	3	5	10	17	19	13	9	3	4		
	PrecMax	139	9	6	8	15	23	27	49	72	30	20	15	18		
	PrecMin	40	0	0	0	0	0	0	1	4	2	1	1	1		
	PrecStd	30	2	2	2	3	5	8	11	13	8	5	3	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.5	-8.2	-8.9	-2.2	5.3	11.5	15.5	13.1	6.5	0.2	-3.4	-12.2		
		MCWB	-9.1	-8.7	-9.4	-3.1	3.5	8.7	12.0	10.1	4.8	-0.8	-3.8	-12.5		
	2%	DB	-14.8	-12.0	-11.7	-5.0	2.4	7.9	12.2	10.6	5.0	-1.0	-6.4	-15.0		
		MCWB	-15.2	-12.3	-12.4	-5.8	1.0	5.4	9.3	9.1	3.9	-1.8	-7.0	-15.4		
	5%	DB	-18.4	-15.5	-15.2	-7.1	1.0	6.2	9.2	8.6	4.1	-1.8	-8.4	-16.8		
		MCWB	-18.7	-15.8	-15.6	-7.6	-0.1	4.1	7.2	7.4	3.2	-2.8	-8.8	-17.1		
	10%	DB	-20.7	-18.0	-17.6	-9.5	-0.2	4.6	6.9	6.5	3.3	-2.8	-11.5	-18.9		
		MCWB	-20.8	-18.3	-17.9	-9.9	-1.0	2.8	5.3	5.5	2.5	-3.7	-12.0	-19.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.1	-8.6	-9.4	-2.8	3.5	8.6	12.2	10.6	5.2	-0.5	-3.8	-12.6		
		MCDB	-8.2	-8.1	-9.0	-2.3	5.5	11.4	15.6	12.1	6.0	0.0	-3.6	-12.3		
	2%	WB	-15.3	-12.4	-12.3	-5.7	1.1	5.6	9.5	9.2	4.1	-1.7	-6.9	-15.4		
		MCDB	-14.7	-12.0	-11.7	-5.1	2.1	7.7	11.8	10.6	4.8	-1.1	-6.5	-15.0		
	5%	WB	-18.7	-15.7	-15.5	-7.6	0.0	4.1	7.3	7.5	3.3	-2.7	-8.7	-17.1		
		MCDB	-18.3	-15.4	-15.2	-7.0	0.8	6.1	9.3	8.7	4.1	-2.0	-8.3	-16.8		
	10%	WB	-20.9	-18.3	-17.9	-9.9	-1.0	2.8	5.4	5.6	2.5	-3.6	-11.9	-19.2		
		MCDB	-20.7	-18.1	-17.7	-9.5	-0.2	4.4	6.7	6.4	3.3	-2.8	-11.5	-18.9		
(35) Mean Daily Temperature Range	MDBR		5.0	4.9	5.0	5.8	4.2	3.6	4.0	2.7	2.2	3.4	5.1	4.8		
	5% DB	MCDBR	7.1	7.7	6.9	7.2	4.7	7.3	9.1	5.4	3.6	2.9	8.0	7.4		
		MCWBR	6.9	7.6	6.8	6.9	4.0	5.5	7.0	4.3	3.0	2.9	7.9	7.3		
	5% WB	MCDBR	7.1	7.8	6.7	7.1	4.6	7.1	8.9	5.4	3.4	3.0	8.1	7.4		
		MCWBR	6.9	7.6	6.6	6.8	4.0	5.5	7.0	4.4	2.9	3.1	8.1	7.2		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.200	0.276	0.318	0.340	0.336	0.375	0.373	0.312	0.241	N/A	N/A		
	taud		N/A	1.972	1.997	1.928	1.948	2.127	2.266	2.345	2.462	2.114	N/A	N/A		
	Ebn at Noon		0	504	693	783	819	843	791	742	698	430	0	0		
	Edh at Noon		0	38	85	129	144	125	106	87	57	33	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.32	1.92	4.39	6.34	6.90	5.13	3.05	1.49	0.42	0.00	0.00		
	RadStd		0.00	0.03	0.08	0.18	0.17	0.40	0.42	0.30	0.10	0.06	0.00	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page